HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Performed deep-dive analysis on **transaction data** and **customer behaviour** to identify emerging financial crime risks and control gaps, presenting findings to the Risk Management team.
- Optimised **transaction monitoring scenarios** by refining detection logic and calibrating thresholds, leading to a **15% reduction** in false positive alerts across key monitoring rules.
- Developed and validated **anomaly detection models** using Python and scikit-learn on a dataset of **~1.2M transactions**, achieving an AUC of **0.88** for suspicious activity.
- Designed and implemented automated controls and alerts within the **Oracle FCCM platform** to enhance operational reporting capabilities for suspicious activity.
- Contributed to the continuous improvement of financial crime controls by conducting root cause analysis on investigation outcomes and alert volumes, informing strategy for control enhancement.
- Supported **audit and regulatory engagements** by providing comprehensive documentation of analytics methodologies, model assumptions, and tuning decisions for transaction monitoring controls.
- Containerised analytical solutions using **Docker** and managed deployments via **Jenkins** pipelines on **AWS EC2** instances, ensuring version-controlled and auditable code promotion.

SKILLS

Financial Crime – Financial Crime Operations, AML, Transaction Monitoring, FCCM, KYC, Suspicious Activity Reporting, Customer Risk Assessment, Network Analysis, Segmentation, Behavioural Analysis

Data Science & Analytics – Python, SQL, Statistical Analysis, Data Mining, Exploratory Data Analysis, Machine Learning, Anomaly Detection, Classification, Clustering, Forecasting, Risk Scoring, CRISP-DM, R, Tableau, Power BI, DAX

Programming & Scripting – Python, Java, Spring Boot, SQL, R

Databases & Data Management – Oracle FCCM, PostgreSQL, MySQL, Oracle, Data Warehousing, Data Quality, ETL, Power Query

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Tools & Methodologies – Jira, Confluence, Agile, Kanban, Change Management, GDPR Compliance, Audit Trail Management

PROJECTS

Financial Crime Transaction Monitoring Optimisation

- Engineered Python scripts to extract and analyse **transaction data** from Oracle FCCM, identifying patterns associated with high-risk activities for a financial crime operations team.
- Developed and evaluated a **classification model** using scikit-learn on a dataset of **~950,000 transaction records** to predict suspicious activity, achieving an F1-score of **0.48**.
- Implemented automated reporting of key performance indicators and control effectiveness metrics using **Power BI**, providing actionable insights to compliance stakeholders.

Customer Behavioural Analytics for Risk Scoring

- Designed and implemented a data pipeline using Python and SQL to aggregate and transform customer transaction history and profile data for behavioural analysis.
- Built a **risk scoring model** in Python, incorporating features derived from transaction velocity and geographic data, to segment customers based on potential financial crime exposure.
- Collaborated with business analysts to refine model parameters and present analytical findings, supporting the enhancement of customer risk assessment processes.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)